



Solar ROI Report

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Scenario Summary

SOLAR SIZE

7 kW

BATTERY SIZE

10 kW

PROJECT COST

PKR 1,234,567

Executive Note

This system is estimated to recover its investment in 1 year 11 months. After payback, the customer continues to benefit from monthly electricity savings. Over the warranty period, total estimated savings can reach PKR 6,372,000.

Executive Summary

TOTAL ANNUAL SAVINGS

PKR 637,200

MONTHLY SAVINGS

PKR 53,100

PAYBACK PERIOD

1.94 years

1 year 11 months

WARRANTY SAVINGS

PKR 6,372,000

NET GAIN

PKR 5,137,433

Calculation Breakdown

PRODUCTION

Annual Solar Units

10,080 units

7 x 1440

STEP 1

PRODUCTION

Peak Units Offset

3,240 units

$\min(10 \times 0.9 \times 360, 10,080)$

STEP 2

PRODUCTION

Low Tariff Units

6,840 units

$10,080 - 3,240$

STEP 3

SAVINGS

Low Tariff Savings

PKR 410,400

$6,840 \times \text{PKR } 60$

STEP 4

SAVINGS

Peak Tariff Savings

PKR 226,800

$3,240 \times \text{PKR } 70$

STEP 5

SAVINGS

Total Annual Savings

PKR 637,200

$\text{PKR } 410,400 + \text{PKR } 226,800$

STEP 6

RETURN

Payback Period

1.94 years

$\text{PKR } 1,234,567 \div \text{PKR } 637,200$

STEP 7

RETURN

Lifetime Savings

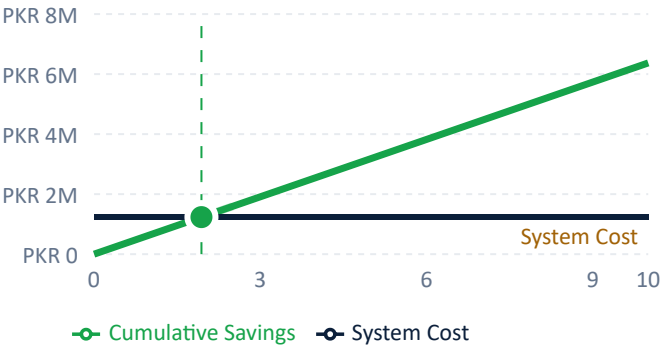
PKR 6,372,000

$\text{PKR } 637,200 \times 10$

STEP 8



Break-even Curve



Assumptions Used

PEAK TARIFF

PKR 70 / unit

LOW TARIFF

PKR 60 / unit

SOLAR PRODUCTION FACTOR

1,440 units / kW / year

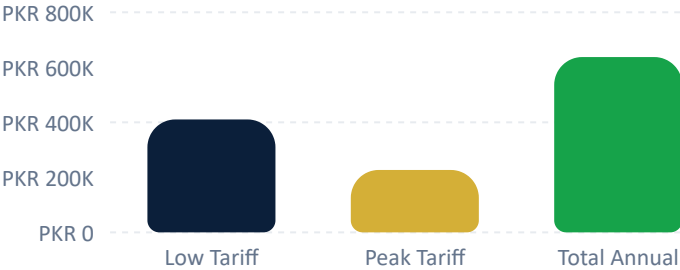
BATTERY OFFSET FACTOR

0.9 units / day / kW battery

WARRANTY YEARS

10 years

Annual Savings Breakdown



Savings Mix

